Document ID: TR-011
Revision: 1.0
Related Procedure: TP-011: Ground Control Station Functional Verification Procedure
Verification Event: VE-006
Verification Method: Test
Date: 18 July 2026
Test Engineer: Serena Green

1. Purpose
Document the results of executing TP-011 to verify that the TOMTOM Ground Control Station satisfies the allocated software verification requirements.

2. Referenced Documents

- TP-011 Verification Procedure
- Verification Workbook
- Applicable Requirements Specification
- Applicable Interface Control Document
- System Requirements Specification

3. Configuration Baseline

Item	**Configuration**
Computer	ASUS G73S
Operating System	Linux Mint 22.3 Cinnamon (64- bit)
Software Version	QGroundControl: V5.0.3-1109-g098330170 64-bit ATAK Launch:
Firmware Version	N/A - Standalone GCS Verification Test
Git Revision	N/A
Procedure Revision	TP-011 Rev. 002
Vehicle Configuration	Not Connected (GCS Standalone Test)
Telemetry Radio	N/A- Not Installed
Date	18 July 2026

4. Test Personnel

Name	**Role**
Serena Green	Test Engineer

Alex Green	Observer

5. Test Results

Verification Item	Expected	Actual	Pass/Fail
Configuration Baseline	Configuration baseline is accurately recorded prior to testing.	Configuration baseline was accurately recorded prior to beginning any verification work or testing.	Pass
Enclosure Integrity Inspection	Enclosure is free of structural damage, loose hardware, or defects.	N/A	N/A- Finalizing GCS preliminary verification.
Hinges, Latches, Lid Stays, and Handles	Hinges, latches, lid stays, and handles operate normally and securely.	N/A	N/A- Finalizing GCS preliminary verification.
Ground Control Station Power-On	Ground Control Station powers on successfully.	Ground Control Station powered on successfully.	Pass
Boot Time	Operating system boots within the required time without errors.	Completely booted in 02:00:49	Pass
Operating System Startup	Operating system loads successfully and reaches a usable desktop environment.	Linux Mint loaded successfully	Pass
QGroundControl Launch	QGroundControl launches successfully	QGroundControl successfully launched and	Pass

	without errors.	repopulated previously saved data without errors.	
ATAK Launch (if installed)	ATAK launches successfully without errors.	ATAK successfully launched following installation and certificate enrollment. The client authenticated with OpenTAKServer using generated PKI certificates and established a secure connection without application errors.	Pass
Display Operation	Display operates normally with a clear and stable image.	Display operating normally with a clear and stable image. Zero anomalies or deviations observed.	Pass
Keyboard and Pointing Device Operation	Keyboard and pointing device respond correctly to user input.	Keyboard and USB mouse both work completely and successfully.	Pass
USB Port Verification	USB ports detect and communicate with compatible devices successfully.	USB devices tested: Mouse, USB Flash Drive. Both detected successfully.	Pass
Ethernet Interface Verification	Ethernet interface is detected and capable of establishing a network	Ethernet interface was successfully detected and established a stable network	Pass

	connection.	connection. Network connectivity was verified through successful access to OpenTAKServer, web-based services, and required software configuration activities.	

6. Requirement Verification Summary

Requirement	Result	Evidence
SYS-GCS-001	PASS	Ground Control Station Power-On; Figure B-1
SYS-GCS-002	PASS	Boot Time; Operating System Startup; Figure B-1
SYS-GCS-003	PASS	QGroundControl Launch; Figure B-2
SYS-GCS-004	PASS	Ethernet Interface Verification; Figure B-3
SYS-GCS-005	PASS	USB Port Verification; Keyboard and Pointing Device Operation; Figure B-2
SYS-GCS-006	PASS	Configuration Baseline; TP-011 Test Results
SYS-HMI-001	PASS	Display Operation; Figure B-2
SYS-HMI-002	PASS	Keyboard and Pointing Device Operation; Figure

		B-2
SYS-HMI-003	PASS	QGroundControl Launch; Figure B-2
SYS-HMI-004	PASS	Map Display; Figure B-2
SYS-HMI-005	PASS	ATAK Launch; Figure B-3
SYS-HMI-006	PASS	Operator Interface Verification; QGroundControl and ATAK operation; Figures B-2 and B-3

7. Deviations

No procedural deviations were encountered during execution of TP-011. Verification items related to the final Ground Control Station enclosure were documented as Not Applicable for this preliminary verification and will be verified during final integrated hardware testing.

8. Observations

The Ground Control Station operated reliably throughout the verification activity. Linux Mint booted successfully, QGroundControl launched without errors, and previously saved configuration data was retained following startup. USB peripherals were detected and functioned as expected. Network connectivity was successfully established through the Ethernet interface, supporting subsequent OpenTAKServer configuration and ATAK client integration. No unexpected system instability, software crashes, or abnormal hardware behavior were observed during testing.

9. Discrepancies

No discrepancies affecting verification were identified during execution of TP-011. Verification items designated as Not Applicable were outside the scope of this preliminary Ground Control Station verification and will be evaluated following completion of the final integrated hardware configuration.

10. Overall Assessment

The Ground Control Station successfully satisfied the functional verification objectives defined in TP-011 for the configured test environment. All completed verification activities met their acceptance criteria, demonstrating successful system startup, operating system functionality, operator interface operation, peripheral detection, network connectivity, and software operation. Verification items associated with the final integrated hardware configuration were deferred until completion of the production Ground Control Station assembly. Based on the results of this verification, the Ground Control Station is considered to have successfully completed preliminary functional verification and is suitable for continued subsystem integration and subsequent verification activities.

11. Evidence References

Figure	Description
Figure B-1	Linux Mint desktop after successful system startup
Figure B-2	QGroundControl operational on Ground Control Station
Figure B-3	OpenTAKServer dashboard accessed via Ethernet connection

12. Approval

Role	Name	Signature	Date
Test Engineer	Serena Green	Serena Green	18 July 2026
Reviewer			

Appendix B: Verification Evidence

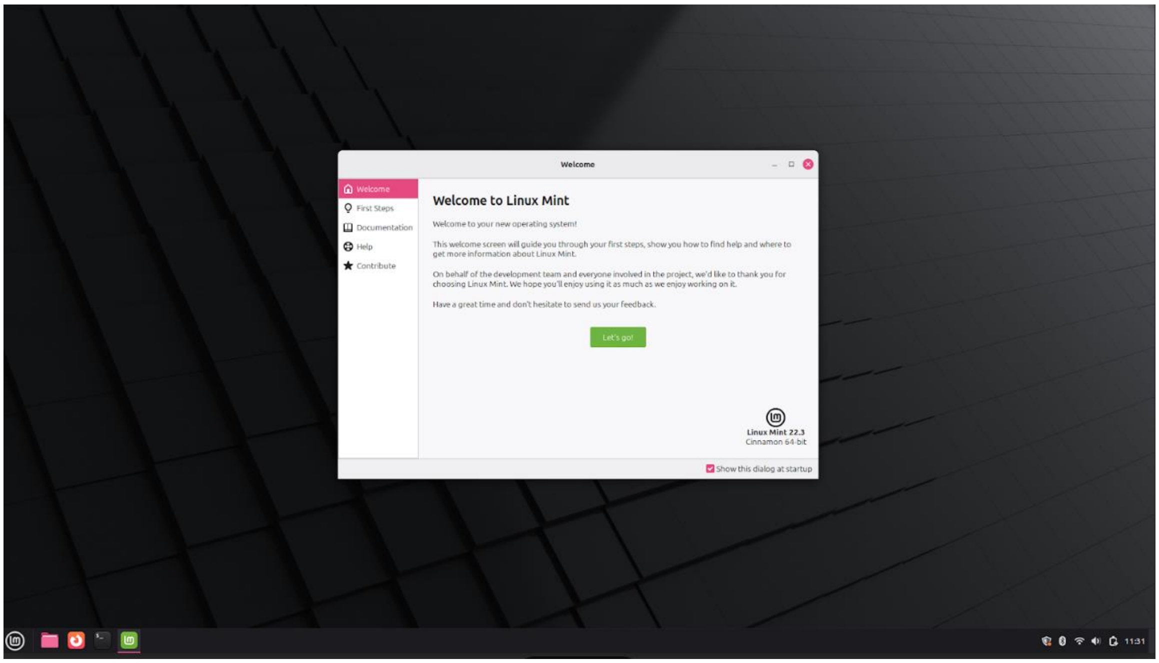

Figure 1-B. *Linux Mint desktop successfully loaded following system startup, verifying successful operating system boot and initialization.*

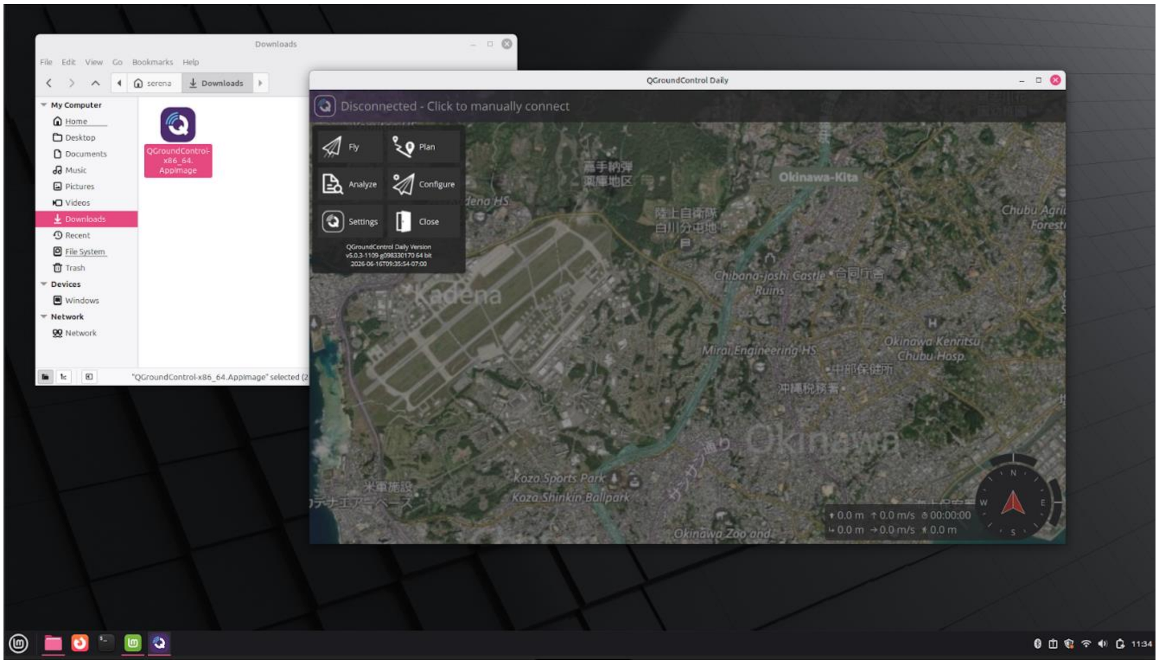

Figure 2-B. *QGroundControl launched successfully on the Ground Control Station with map data displayed and the application operating normally.*

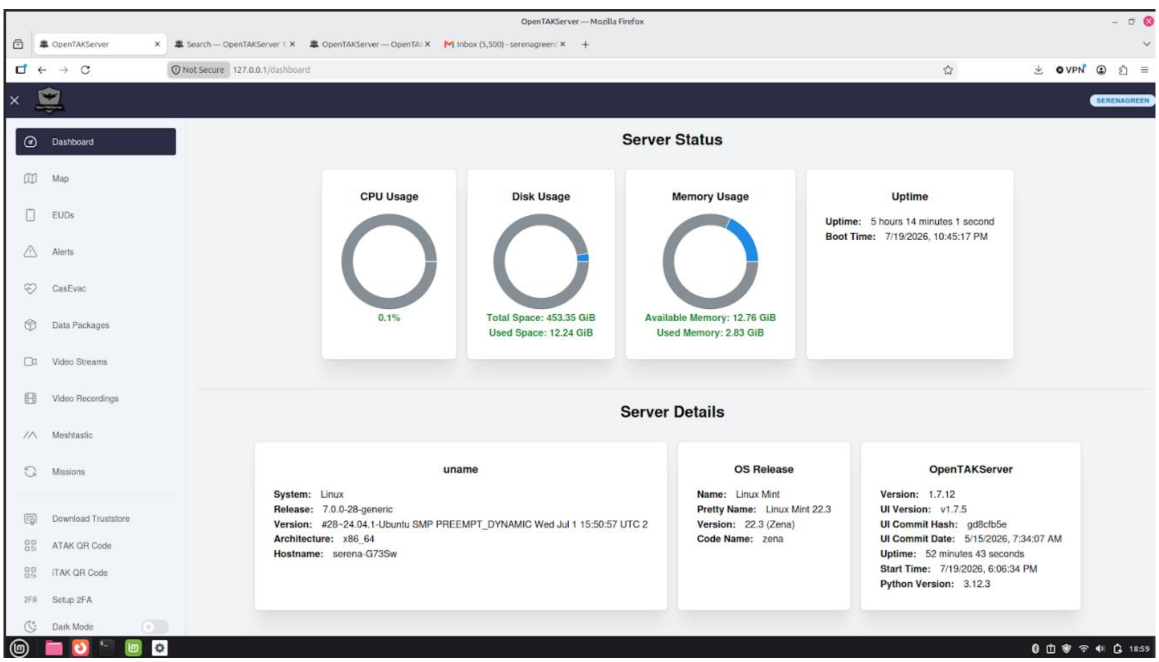

Figure 3-B. *OpenTAKServer web interface successfully accessed through the Ethernet network connection following server configuration, demonstrating functional network connectivity and software operation.*